

ENVY-FREE CHORE ALLOCATION WITH UNIT SUBSIDIES UNDER NEGATIVE DICHOTOMOUS VALUATIONS

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Abstract

We study the allocation of indivisible chores with subsidies under negative dichotomous valuations, where the marginal disutility of every chore is either zero or one. This is the chore analogue of the dichotomous goods model of Barman et al. [BKNS22], for which a subsidy of at most one unit per agent is known to suffice for envy-freeness.

We show that the same optimal guarantee holds for chores, under no structural assumption beyond binary marginals. For every instance with negative dichotomous valuations, there exists a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is envy-free. Moreover, the allocation $\mathcal{A}$ is EF1 even before subsidies are paid. The bound is tight, and the allocation and subsidies can be computed in polynomial time in the value-oracle model.

Our algorithm starts from an envy-free partial allocation produced by the algorithm of Tao et al. [TWYZ25], assigns the remaining chores to distinct agents in a tail strongly connected component selected at termination, and determines the subsidised agents through a backward closure in the associated equality graph.

We extend the guarantee to objective signed dichotomous valuations, in which every item is a good for every agent or a chore for every agent, by showing that the one-good extension argument of Barman et al. remains valid when the bundles already contain chores. We show that unit subsidies are compatible with Pareto optimality under additivity but not in general. Finally, chores for every agent can be added to any envy-free solution with unit subsidies at the cost of at most one more unit per agent. This gives a goods-first algorithm with at most two units per agent for objective signed valuations, and reduces the same guarantee for doubly monotone dichotomous valuations to a single insertion property.

1 Introduction

Discrete fair division studies the allocation of indivisible resources among agents with heterogeneous preferences, and lies at the interface of mathematical economics and computer science [BCE⁺16, End17]. A central fairness criterion is *envy-freeness* [Fol67, Var74]: no agent prefers another agent’s bundle to her own. For indivisible items, however, an envy-free

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allocation need not exist, as the standard one-item two-agent instance illustrates. This has motivated a large literature on relaxations of envy-freeness, including envy-freeness up to one good (EF1) [Bud11, LMMS04] and the stronger notion of envy-freeness up to any good (EFX) [CKM⁺19].

A second and complementary line of work retains exact envy-freeness by allowing monetary subsidies. In addition to her allocated bundle A_i , each agent i receives a subsidy $p_i \geq 0$ from a third party, and the outcome is envy-free if

$$v_i(A_i) + p_i \geq v_i(A_j) + p_j \quad \text{for all } i, j.$$

Halpern and Shah [HS19] show that every instance admits an envy-freeable allocation and study the minimum subsidy required to achieve envy-freeness. Under the standard normalization that each item value is at most one,¹ they show that, for a given envy-freeable allocation, a total subsidy of at most $(n - 1)m$ may be required in the worst case, and conjecture that a total subsidy of $n - 1$ always suffices when the allocation can also be chosen. Brustle et al. [BDN⁺20] prove this conjecture for additive valuations, obtaining an allocation that is also EF1 and balanced, and show that for general monotone valuations a subsidy of at most $2(n - 1)$ per agent suffices.

Barman et al. [BKNS22] obtain a substantially stronger guarantee for dichotomous valuations. When every marginal $v_i(S \cup \{g\}) - v_i(S)$ belongs to $\{0, 1\}$, they show that there exists an allocation admitting subsidies in $\{0, 1\}^n$, with total subsidy at most $n - 1$, and that such an outcome can be computed in polynomial time in the value-oracle model. This bound is tight: for n agents and a single good valued at one by every agent, the $n - 1$ agents who do not receive the good each require one unit of subsidy.

In this paper, we consider the corresponding problem for indivisible chores, where receiving an additional item weakly decreases an agent’s utility. Chore-allocation problems arise naturally in applications such as shift assignment, on-call duties, teaching and refereeing loads, committee service, and maintenance tasks. Monetary compensation is also natural in such settings, where it may be interpreted as compensation for undertaking an undesirable task or additional workload.

We study *negative dichotomous* valuations: for every agent i , every $S \subseteq M$, and every $g \in M \setminus S$,

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, -1\}.$$

Thus, every item is a chore for every agent, and adding a chore changes an agent’s utility by either zero or minus one. This domain is the sign analogue of the dichotomous goods model of [BKNS22].

Binary marginals capture settings in which an additional task either creates a new unit of burden or can be absorbed by an existing commitment. For example, an agent already covering an on-call window may incur no additional cost from another incident during that window, or a driver already routed through a district may absorb an additional stop at no extra cost. Since whether a chore is costly may depend on the rest of the assigned bundle, the model naturally allows non-additive valuations, including valuations that are not subadditive.

¹Such a normalization is necessary when stating absolute bounds on monetary subsidies.

The baseline and the lower bound. Two observations provide useful benchmarks for the problem, and we state them explicitly so that the contribution can be measured against them. On the one hand, the existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS⁺25, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies directly and yields a subsidy of at most $n - 1$ per agent and $n(n - 1)/2$ in total [KMS⁺25, Theorem 1]. Two details concerning this application are worth noting. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form of Definition 5, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS⁺25, §2] cites [LMMS04] and [BSV21]. In particular, while [LMMS04] concerns goods, [BSV21] provides the corresponding existence result required in the chore setting.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n - 1$ lower bound transfers). Take n agents and $m = n - 1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n - 1$ agents hold one chore each and one agent holds nothing, each of the $n - 1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n - 1$. No allocation does better: consider any complete allocation $\mathcal{A} = (A_1, \dots, A_n)$, and write $s_i = |A_i|$. Since there are only $n - 1$ chores, $\sum_{i=1}^n s_i = n - 1$, and therefore at least one agent, say h , receives no chore: $s_h = 0$. Suppose that a subsidy vector $p \in \mathbb{R}_+^n$ makes the allocation envy-free. In cost form, envy-freeness requires

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j.$$

Comparing every agent i with the empty-bundle agent h , we obtain

$$s_i - p_i = c_i(A_i) - p_i \leq c_i(A_h) - p_h = -p_h.$$

Hence

$$p_i \geq s_i + p_h \geq s_i \quad \text{for every } i.$$

Summing over all agents gives

$$\sum_{i=1}^n p_i \geq \sum_{i=1}^n s_i = n - 1.$$

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Theorem 2 (Main result). *For every instance with negative dichotomous valuations, there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0, 1\}^n$, hence with total subsidy at most $n - 1$; the allocation $\mathcal{A}$ is moreover EF1 on its own, before any subsidy is paid; and such an allocation and subsidy vector can be computed in polynomial time given value-oracle access.*

Example 1 shows this bound is tight. The guarantee of [KMS⁺25] is $n - 1$ per agent and $n(n - 1)/2$ in total, whereas our target is at most one per agent and $n - 1$ in total, yielding a linear-factor improvement.

Our contribution. We establish the statement above, for every number of agents (Theorem 22). The allocation produced is EF1 in the sense of Definition 6 before any money changes hands, and the addition of subsidies upgrades the resulting allocation to exact envy-freeness. The proof takes as input an envy-free *partial* allocation — the algorithm of Tao et al. [TWYZ25] produces one in polynomial time, leaving at most $n - 1$ chores unassigned — and completes it: the residual chores are assigned to distinct agents inside the strongly connected component at which that algorithm halts, and the paid set is the backward closure of those recipients under the equality relation $c_i(X_i) = c_i(X_j)$. Envy-freeness of the completion is verified directly, so no appeal to the characterisation of envy-freeable allocations is required. Section 2 fixes the model and records what transfers from the goods theory unchanged; Section 3 contains the proof.

Further results. Section 4 extends the unit-subsidy guarantee to objective signed dichotomous valuations, in which every item is a good for every agent or a chore for every agent. The chores are allocated first by Theorem 22, and the goods are then inserted one at a time by the one-good extension of Barman et al. [BKNS22], whose proof remains valid when the bundles already contain chores (Theorem 26). Section 5 shows that unit subsidies are compatible with Pareto optimality for binary additive costs, and more generally for additive objective signed dichotomous valuations, but not for general negative dichotomous valuations, even with two agents and identical binary submodular costs. Section 6 reverses the order of allocation, which is the order that doubly monotone valuations require. Items that are chores for every agent can be added to any envy-free solution with unit subsidies at the cost of at most one more unit per agent and $n - 1$ more in total (Lemma 34). This gives a goods-first algorithm for objective signed valuations with at most two units per agent and $2(n - 1)$ in total, and it reduces the same guarantee for doubly monotone dichotomous valuations to a single insertion property, which remains open for items that are a good for some agents and a chore for others.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n - 1$, Goko et al. [GIK⁺24] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The work of Kawase et al. [KMS⁺25] is particularly relevant to the present setting because it handles general doubly monotone valuations, and hence includes non-additive chore

valuations. Working with doubly monotone valuations under the two-sided normalisation, it separates the allocation and subsidy steps that are handled jointly in [BDN⁺20]: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n - 1$ per agent and $n(n - 1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, this yields the stated subsidy guarantee throughout that class. This result, therefore, provides the natural benchmark against which we compare our subsidy bound.

A complementary approach dispenses with subsidies and instead considers approximate notions of envy-freeness. Bu et al. [BSY23] show that EFX allocations always exist and can be computed in polynomial time for dichotomous goods valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. These results provide two complementary guarantees on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. This paper establishes the first branch for chores.

The chores side. Subsidies have also been studied in the allocation of indivisible chores, although much of this literature concerns fairness notions other than envy-freeness. Subsidies have, in particular, been studied for achieving proportionality under additive costs. For agents with equal entitlements, tight subsidy bounds are known: the optimal total subsidy is $n/4$ when n is even and $(n^2 - 1)/(4n)$ when n is odd [WXZ25]. In the more general weighted setting, a total subsidy of $n/3 - 1/6$ is known to suffice [WZ24], and Garg et al. [GSW25] obtain this guarantee together with Pareto optimality. These results concern proportionality rather than envy-freeness, and therefore do not directly bound the subsidy required in our setting.

Turning to approximate fairness without subsidies, there is a direct chore analogue of [BSY23]. Tao et al. [TWYZ25] study chores with binary marginals and show that for binary *additive* costs, an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half pays 0 or 1 per agent and obtain exact envy-freeness, is what this paper establishes. The restriction to additive costs there cannot simply be dropped: Lin et al. [LLTZ26] exhibit instances with 18 agents and 53 chores, one with binary XOS costs and one with binary supermodular costs, in which no complete EFX allocation exists at all; the binary submodular case remains open. Thus, complete EFX allocations need not exist throughout the binary-marginal chore domain, whereas the allocation constructed below is always EF1, and becomes exactly envy-free once at most one unit of subsidy per agent is paid. Exact envy-freeness without subsidies is not always attainable; indeed, Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Recent work has also considered subsidies for additive instances involving both goods and chores. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN⁺20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \leq 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n - 1$ in total. In particular, this establishes the optimal subsidy bound for additive chore instances.

This result does not subsume the present setting, for the same reason that [BKNS22] is not a corollary of [BDN⁺20]: dichotomous valuations need not be additive, and additive valuations need not be dichotomous. The relationship among these results can be summarised as follows:

	goods	chores
additive	[BDN ⁺ 20]	[LMS26]
dichotomous	[BKNS22]	this paper

each of the four yielding a per-agent subsidy bound of one and a total subsidy bound of $n - 1$. Note also that [LMS26] does not subsume [KMS⁺25] for the valuation class considered here: for valuations that are dichotomous but not additive, the $n - 1$ per agent of [KMS⁺25] was the best previously known bound, and the present result improves this by a linear factor.

To the best of our knowledge, envy-freeness with subsidies has not previously been studied for chores under dichotomous costs.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i \in N} \rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \dots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i . The allocation is *complete* if $\bigcup_{i \in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \dots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\text{SW}(\mathcal{A}) = \sum_{i \in N} v_i(A_i)$.

Definition 3 (Envy-Freeness). *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is envy-free (EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution). *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ and a subsidy vector $p = (p_1, \dots, p_n) \in \mathbb{R}_+^n$ constitute an envy-free solution $(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}_+^n$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $M(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will also use the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS⁺25], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 5 (EF1, two-sided form). *An allocation $\mathcal{A}$ is envy-free up to one item (EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

Every instance in this paper is an all-chores instance, and there Definition 5 takes a simpler form: the item removed can always be taken from the envious agent's own bundle. We record that specialisation, since it is the form the results below establish.

Definition 6 (EF1 for chores). *A complete allocation $\mathcal{A}$ of chores is EF1 iff for all $i, j \in N$, either i does not envy j , that is $c_i(A_i) \leq c_i(A_j)$, or there is a chore $e \in A_i$ with*

$$c_i(A_i \setminus \{e\}) \leq c_i(A_j).$$

Definition 6 is exactly Definition 5 on an all-chores instance. Taking $X = \emptyset$ in the first case and $X = \{e\} \subseteq A_i$ in the second gives one direction; for the converse, a removed item lying in A_j only lowers $c_i(A_j)$'s counterpart on the wrong side of the inequality, so nothing is gained by removing from the envied bundle when every item is a chore.

2.1 Negative Dichotomous Valuations

Definition 7 (Negative Dichotomous Valuation). *A valuation $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$ is negative dichotomous iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \quad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \quad \text{so} \quad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

Definition 8 (Dichotomous cost function). *A set function $c : 2^M \rightarrow \mathbb{R}$ is dichotomous if $c(\emptyset) = 0$ and every marginal $c(S \cup \{g\}) - c(S)$ lies in $\{0, 1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0, 1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly

the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j , read as “my own burden net of my compensation is no worse than what I perceive of yours”. We use whichever of v_i and c_i is clearer locally and always state which.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) := v_i(A_j) - v_i(A_i) = c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j . The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j) \in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i , the empty path of weight 0 included.

The following two results underpin much of the literature on subsidies for envy-freeness. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 9 ([HS19]). *For any allocation $\mathcal{A} = (A_1, \dots, A_n)$, the following are equivalent.*

- (i) $\mathcal{A}$ is envy-freeable.
- (ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N , $\sum_{i \in N} v_i(A_i) \geq \sum_{i \in N} v_i(A_{\sigma(i)})$.
- (iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 10 ([HS19]). *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p_i^* := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p_i^* \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 9 has the standing consequence that an envy-freeable allocation can always be obtained from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. The central question is therefore not the existence of an envy-freeable allocation, but the amount of subsidy guaranteed by Theorem 10.

Specialising in our model gives the following, which we use throughout.

Observation 11 (Integrality). *Let $\{v_i\}_{i \in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\leq 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}_{\geq 0}^n$.*

Proof Fix i and $S = \{g_1, \dots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 7; hence $v_i(S) \in \mathbb{Z}$ and $-k \leq v_i(S) \leq 0$. Consequently each arc weight $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 10, $p_i^* = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i , and it is non-negative because the empty path has weight 0. $\square$

Observation 11 is what makes the target statement — a subsidy vector in $\{0, 1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p_i^* \leq 1$ for every i is the same as showing $p^* \in \{0, 1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

The following observation further simplifies the target guarantee.

Proposition 12 (Some agent is always unpaid). *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p_h^* = 0$. Consequently, if $p^* \in \{0, 1\}^n$ then $\sum_{i \in N} p_i^* \leq n - 1$.*

Proof By Theorem 9(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h , and P followed by Q is a walk from P 's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n - 1$ coordinates of p^* can equal 1. $\square$

Proposition 12 justifies the word “hence” in the target statement of Section 1: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. For the pointwise-minimal subsidy the target is therefore the single-clause statement

$$\exists \mathcal{A} : \max_{i \in N} \ell_{\mathcal{A}}(i) \leq 1.$$

3 Unit Subsidies for Negative Dichotomous Valuations

This section proves the main theorem: for every number of agents n and every negative dichotomous instance there is an allocation admitting an envy-free solution with a subsidy of 0 or 1 per agent, at most $n - 1$ in total, computable in polynomial time. Beyond Definition 8 nothing is assumed of the cost functions — not additivity, not submodularity, not any relation between different agents’ cost functions — and the number of chores is unrestricted.

The argument takes as input an envy-free *partial* allocation produced by the algorithm of Tao, Wu, Yu and Zhou [TWYZ25], and completes it. That algorithm, its Theorem 5.1, and the three update rules recorded in Section 3.1 are due to [TWYZ25] and are restated here for reference; the completion of Section 3.2 onwards, and every lemma and theorem after Lemma 15, are ours.

We work in cost form throughout, so an envy-free solution (Definition 4) is the requirement

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N, \quad (1)$$

and a partial allocation is envy-free (Definition 3) when

$$c_i(A_i) \leq c_i(A_j) \quad \text{for all } i, j \in N. \quad (2)$$

For $S \subseteq M$ and $e \in M \setminus S$ write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal, and for a partial allocation $\mathcal{A}$ write $R(\mathcal{A}) := M \setminus \bigcup_{i \in N} A_i$ for its *residue*.

Observation 13. For every $i \in N$ and all $S, T \subseteq M$,

$$c_i(S) < c_i(T) \implies c_i(S) \leq c_i(T) - 1.$$

Moreover $c_i(S) \leq c_i(T)$ whenever $S \subseteq T$.

Proof By Observation 11 every c_i is integer valued, so a strict inequality between two of its values is a gap of at least 1. For monotonicity, enumerate $T \setminus S = \{f_1, \dots, f_q\}$ and telescope: $c_i(T) - c_i(S) = \sum_{u=1}^q c_i(f_u \mid S \cup \{f_1, \dots, f_{u-1}\})$, a sum of terms in $\{0, 1\}$ by Definition 8, hence non-negative. $\square$

3.1 The terminal state of the partial-allocation algorithm

Theorem 14 ([TWYZ25], Theorem 5.1). For every n and all dichotomous cost functions $c_1, \dots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free in the sense of (2) and satisfies $|R(X)| \leq n - 1$.

The proof below uses not only the statement of Theorem 14 but the state in which the algorithm of [TWYZ25] halts, so we record that algorithm. It maintains a partial allocation $X = (X_1, \dots, X_n)$, envy-free throughout, together with the *equality graph*

$$G = (N, E), \quad (i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j), \quad (3)$$

and repeats the following three rules in the stated order, halting only when none applies.

- (R1) If some $e \in R(X)$ and some $i \in N$ satisfy $c_i(e \mid X_i) = 0$, assign e to i .
- (R2) If some $e \in R(X)$ and some arc $(i, j) \in E$ lying on a directed cycle of G satisfy $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i .
- (R3) Otherwise select a strongly connected component S of G with no arc of E leaving S . If $|R(X)| \geq |S|$, assign one residual chore to each agent of S ; otherwise halt.

The loop runs while $R(X) \neq \emptyset$, so the only exit with $R(X) \neq \emptyset$ is the final clause of (R3).

For the remainder of this section fix the following data: let X be the partial allocation at which the algorithm halts, let $R := R(X)$ and $r := |R|$, and, when $r \geq 1$, let S be the component selected by (R3) at the halting step, with G and E as in (3) evaluated at that state.

Lemma 15. Suppose $r \geq 1$. Then

- (a) $r < |S|$;
- (b) $c_i(e \mid X_i) = 1$ for every $i \in N$ and every $e \in R$;

(c) $c_i(e \mid X_j) = 1$ for every arc $(i, j) \in E$ with $i, j \in S$ and every $e \in R$;

(d) $c_i(X_i) < c_i(X_j)$ for every $i \in S$ and every $j \in N \setminus S$.

Proof Since $r \geq 1$, the algorithm halted at the final clause of (R3); hence neither (R1) nor (R2) applied at the halting state, and $|R| < |S|$, which is (a).

(b) If $c_i(e \mid X_j) = 0$ for some $i \in N$ and some $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$ by Definition 8, so the value is 1.

(c) Let $(i, j) \in E$ with $i, j \in S$. Since S is strongly connected there is a directed path in G from j to i ; taking one with no repeated vertex and appending the arc (i, j) yields a directed cycle of G through (i, j) . If $c_i(e \mid X_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and that cycle. Hence the marginal is not 0, so it is 1.

(d) Let $i \in S$ and $j \in N \setminus S$. No arc of E leaves S , so $(i, j) \notin E$, i.e. $c_i(X_i) \neq c_i(X_j)$ by (3). Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_j)$, so the inequality is strict. $\square$

3.2 The completion

Assume $r \geq 1$ and write $R = \{e_1, \dots, e_r\}$. By Lemma 15(a), $r < |S|$, so we may fix r distinct agents

$$T = \{t_1, \dots, t_r\} \subseteq S,$$

chosen arbitrarily. Define $\mathcal{A} = (A_1, \dots, A_n)$ by

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \ (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases} \quad (4)$$

The t_k are distinct and each chore of R is used exactly once, so the bundles of $\mathcal{A}$ are pairwise disjoint and cover M : $\mathcal{A}$ is a complete allocation. Members of T are called *recipients*.

Definition 16 (Subsidy set). *Let $P \subseteq N$ be the smallest set satisfying*

(P1) $T \subseteq P$; and

(P2) *if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.*

Equivalently, P is obtained by initialising $P := T$ and repeatedly adjoining any $i \in N \setminus S$ having an arc of E into the current set, until no further agent can be adjoined; the process terminates because N is finite and P only grows. Define

$$p_i := \begin{cases} 1, & i \in P, \\ 0, & i \in N \setminus P. \end{cases} \quad (5)$$

The complete procedure is summarised in Algorithm 1. The partial-allocation phase in Line 2 is Algorithm 3 of [TWYZ25]; our contribution is the completion and subsidy construction in the subsequent lines.

Algorithm 1 EF1 allocation with at most one unit of subsidy per agent for negative dichotomous valuations

Require: Agents $N = [n]$, chores M , and value-oracle access to negative dichotomous valuations $\{v_i\}_{i \in N}$.

Ensure: A complete allocation $\mathcal{A}$ that is EF1, together with a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and $\sum_{i \in N} p_i \leq n - 1$.

```

1: Set  $c_i = -v_i$  for every  $i \in N$ .
2: Run Algorithm 3 of [TWYZ25] on  $\{c_i\}_{i \in N}$ .
3: Let  $X = (X_1, \dots, X_n)$  be its terminal partial allocation and let  $R = M \setminus \bigcup_{i \in N} X_i$ .
4: if  $R = \emptyset$  then
5:   return  $(X, 0)$ .
6: Let  $S$  be the tail strongly connected component selected at the terminating application
   of Rule (R3).
7: Write  $R = \{e_1, \dots, e_r\}$ .
8: Choose arbitrary distinct agents  $T = \{t_1, \dots, t_r\} \subseteq S$ .  $\triangleright r < |S|$  by Lemma 15(a)
9: for  $k = 1, \dots, r$  do
10:    $A_{t_k} \leftarrow X_{t_k} \cup \{e_k\}$ .
11: for  $i \in N \setminus T$  do
12:    $A_i \leftarrow X_i$ .
13: Construct the equality graph  $G = (N, E)$  on the agents, where
       $(i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j)$ .
14: Initialise  $P \leftarrow T$ .
15: while there exist  $i \in N \setminus S$  and  $j \in P$  with  $(i, j) \in E$  and  $i \notin P$  do
16:    $P \leftarrow P \cup \{i\}$ .
17: for  $i \in N$  do
18:    $p_i \leftarrow \mathbf{1}[i \in P]$ .
19:  $\mathcal{A} \leftarrow (A_1, \dots, A_n)$ .
20: return  $(\mathcal{A}, p)$ .
```

Lemma 17. $P \setminus T \subseteq N \setminus S$: every agent of P that is not a recipient lies outside S .

Proof Let $P_0 := T$ and let P_{u+1} be P_u together with every $i \in N \setminus S$ having an arc of E into P_u , so that $P = \bigcup_{u \geq 0} P_u$. We show $P_u \setminus T \subseteq N \setminus S$ by induction on u . For $u = 0$ the set $P_0 \setminus T$ is empty. For the step, every agent adjoined in passing from P_u to P_{u+1} lies in $N \setminus S$ by construction. $\square$

Lemma 18. $p \in \{0, 1\}^n$ and $\sum_{i \in N} p_i = |P| \leq n - 1$.

Proof That $p \in \{0, 1\}^n$ with $\sum_i p_i = |P|$ is immediate from (5). By Lemma 17, $P \subseteq T \cup (N \setminus S)$, and this union is disjoint because $T \subseteq S$; hence $|P| \leq |T| + |N \setminus S| = r + n - |S|$. By Lemma 15(a), $r \leq |S| - 1$, so $|P| \leq n - 1$. $\square$

Lemma 19. Let $i \in S$, let $t \in T$ and let $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

Proof Note $t \in T \subseteq S$. Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_t)$.

If $c_i(X_i) < c_i(X_t)$, then $c_i(X_t) \geq c_i(X_i) + 1$ by Observation 13, and $c_i(X_t \cup \{e\}) \geq c_i(X_t)$ by monotonicity, again Observation 13.

Otherwise $c_i(X_i) = c_i(X_t)$. If $i = t$ then $c_i(X_t \cup \{e\}) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$ by Lemma 15(b). If $i \neq t$ then $(i, t) \in E$ by (3), with $i, t \in S$, so $c_i(e \mid X_t) = 1$ by Lemma 15(c) and therefore $c_i(X_t \cup \{e\}) = c_i(X_t) + 1 = c_i(X_i) + 1$. $\square$

3.3 Envy-freeness of the completion

Theorem 20. Let X, R, S be as in Section 3.1 with $r \geq 1$, let $\mathcal{A}$ be as in (4) and let p be as in (5). Then $(\mathcal{A}, p)$ is an envy-free solution.

Proof Fix $i, j \in N$; we verify (1), that is $c_i(A_i) - p_i \leq c_i(A_j) - p_j$. Throughout, $X_j \subseteq A_j$ for every j by (4), so by monotonicity (Observation 13) and (2),

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i). \quad (6)$$

Note also that $T \subseteq P$, so $i \notin P$ implies $i \notin T$ and hence $A_i = X_i$; likewise $j \notin P$ implies $A_j = X_j$ and $p_j = 0$. Since N is the disjoint union of $T, P \setminus T$ and $N \setminus P$, the following three cases are exhaustive.

Case 1: $i \in T$. Then $p_i = 1$ and, by Lemma 15(b), $c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$, so $c_i(A_i) - p_i = c_i(X_i)$. Also $i \in T \subseteq S$.

(1a) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$, so $c_i(A_j) \geq c_i(X_i) + 1$ by Lemma 19. Hence $c_i(A_j) - p_j \geq c_i(X_i)$.

(1b) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$. By Lemma 17, $j \notin S$; since $i \in S$, Lemma 15(d) gives $c_i(X_i) < c_i(X_j)$, hence $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 13. Hence $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

(1c) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

Case 2: $i \in P \setminus T$. Then $p_i = 1$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i) - 1$. For every j we have $p_j \leq 1$ and $c_i(A_j) \geq c_i(X_i)$ by (6), hence $c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i$.

Case 3: $i \in N \setminus P$. Then $p_i = 0$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i)$.

(3a) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

(3b) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$; we claim $c_i(A_j) \geq c_i(X_i) + 1$. If $i \in S$ this is Lemma 19. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in T \subseteq P$ while $i \notin P$; so $(i, j) \in E$ by (3), and $j \in P$, so (P2) of Definition 16 places $i \in P$ — contradicting $i \notin P$. Hence $c_i(X_i) < c_i(X_j)$, so $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 13, and $c_i(A_j) \geq c_i(X_j)$ by monotonicity. In either case $c_i(A_j) - p_j \geq c_i(X_i)$.

(3c) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$; we claim $c_i(X_j) \geq c_i(X_i) + 1$. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in P$ while $i \notin P$; so $(i, j) \in E$, and (P2) places $i \in P$, a contradiction. Hence $c_i(X_i) < c_i(X_j)$. If instead $i \in S$, then $j \notin S$ by Lemma 17, and Lemma 15(d) gives $c_i(X_i) < c_i(X_j)$. In either case Observation 13 yields $c_i(X_j) \geq c_i(X_i) + 1$, so $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

In every sub-case $c_i(A_j) - p_j \geq c_i(X_i) = c_i(A_i) - p_i$ in Cases 1 and 3, and the inequality was shown directly in Case 2. $\square$

3.4 The EF1 property of the completion

Corollary 21. *The complete allocation $\mathcal{A}$ constructed in Section 3.2 is EF1 in the sense of Definition 6, hence also in the sense of Definition 5. More precisely, every non-recipient $i \in N \setminus T$ is envy-free in $\mathcal{A}$, while for every recipient $t_k \in T$, removing the residual chore e_k assigned to t_k eliminates all of t_k 's envy.*

Proof Recall that the partial allocation $X = (X_1, \dots, X_n)$ is envy-free, and that $X_j \subseteq A_j$ for every $j \in N$. First, let $i \in N \setminus T$. Then $A_i = X_i$. Hence, for every $j \in N$, envy-freeness of X and monotonicity of c_i give

$$c_i(A_i) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Thus i does not envy any agent in $\mathcal{A}$. Now let $i = t_k \in T$. By construction,

$$A_i = X_i \cup \{e_k\},$$

and hence

$$A_i \setminus \{e_k\} = X_i.$$

Therefore, for every $j \in N$,

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j),$$

where the first inequality follows from envy-freeness of X and the second from monotonicity.

Thus, for every recipient, removing the single newly assigned chore from her own bundle eliminates her envy towards every other agent. Consequently, $\mathcal{A}$ is EF1. $\square$

3.5 The main theorem

Theorem 22. *For every n and every instance $\langle N, M, \{v_i\}_{i \in N} \rangle$ with negative dichotomous valuations there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover, the allocation $\mathcal{A}$ is EF1. Such an allocation and subsidy vector can be computed in polynomial time given value-oracle access.*

Proof Pass to cost form, $c_i = -v_i$, which is dichotomous in the sense of Definition 8. Run Algorithm 3 of [TWYZ25] and retain its terminal partial allocation X , the residue R with $r = |R|$. If $r \geq 1$, retain also the tail strongly connected component S selected at the terminating application of rule (R3), all as fixed in Section 3.1. The statement of Theorem 14 alone does not supply S .

If $r = 0$ then X is a complete allocation satisfying (2), so $(X, \mathbf{0})$ is an envy-free solution with $\sum_i p_i = 0 \leq n - 1$. Since every envy-free allocation is EF1, X is also EF1.

If $r \geq 1$, construct T , $\mathcal{A}$ and p as in Section 3.2. Then $\mathcal{A}$ is a complete allocation, Theorem 20 shows $(\mathcal{A}, p)$ is an envy-free solution, and Lemma 18 gives $p \in \{0, 1\}^n$ with $\sum_i p_i \leq n - 1$. The allocation $\mathcal{A}$ is EF1 by Corollary 21.

For the running time, Algorithm 3 of [TWYZ25] is polynomial and yields X , R and the tail component S of its terminating (R3) step if $r \geq 1$. Constructing $\mathcal{A}$ from (4) takes $r \leq n - 1$ assignments. The graph G of (3) has n vertices and is determined by the n^2 values $c_i(X_j)$, one oracle call each. Computing P is a backward reachability search in G from T restricted to vertices outside S , which takes $O(n^2)$ time. Hence the work after Theorem 14 is polynomial, and so is the whole procedure. $\square$

By Example 1 the bound of Theorem 22 is tight: there are instances on which every complete allocation requires total subsidy at least $n - 1$.

Remark 23. Theorem 22 uses Theorem 14 together with the four properties of its halting state recorded in Lemma 15, which are read off the rules (R1)–(R3) of [TWYZ25] restated in Section 3.1; the statement of Theorem 14 alone does not supply them. Envy-freeness is verified directly from (1) in Theorem 20, so the argument does not invoke Theorem 9 or Theorem 10, and the subsidy exhibited need not be the pointwise-minimal one. Part (a) of Lemma 15 is used only in Lemma 18: the conclusion of Theorem 20 holds whenever $r \leq |S|$.

4 Unit Subsidies for Objective Signed Dichotomous Valuations

We now consider mixed instances in which every item is commonly classified as a good or a chore. Such common-sign settings are referred to as *objective* goods and chores or *objective valuations* in the fair-division literature [ACIW22, BLV26]. We additionally impose signed binary marginals.

Definition 24 (Objective signed dichotomous valuations). *A valuation profile $\{v_i\}_{i \in N}$ is objective signed dichotomous if the item set admits a common partition $M = G \dot{\cup} C$ such that, for every agent $i \in N$ and every $S \subseteq M$,*

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, 1\} \quad \text{for every } g \in G \setminus S,$$

and

$$v_i(S \cup \{c\}) - v_i(S) \in \{0, -1\} \quad \text{for every } c \in C \setminus S.$$

Thus, every item is commonly regarded as either a good or a chore, although its marginal value may depend on the bundle to which it is added. In particular, the valuations need not be additive, submodular, or subadditive. The class is a subclass of doubly monotone valuations in which the classification of items into goods and chores is common across agents. Since $v_i(\emptyset) = 0$ and every marginal is an integer, the telescoping argument of Observation 11 shows that every such valuation is integer valued.

The plan is to allocate the chores by Theorem 22 and then the goods by the algorithm of Barman et al. [BKNS22]. That algorithm is incremental: it inserts the goods one at a time, and its only invariant is that the current solution is envy-free with subsidies in $\{0, 1\}^n$. So it can take over from any solution satisfying the invariant, provided its one-good step remains valid when the bundles already contain chores. The following lemma records that step.

Lemma 25. *Let M' be a set of allocated items and let $A = (A_1, \dots, A_n)$ be a complete allocation of M' . Suppose that the valuations are integer-valued and that there exists $p \in \{0, 1\}^n$ such that (A, p) is envy-free. Let $g \notin M'$ satisfy*

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, 1\}$$

for every agent i and every $S \subseteq M'$. Then there exist a complete allocation A' of $M' \cup \{g\}$ and $p' \in \{0, 1\}^n$ such that (A', p') is envy-free. Moreover, (A', p') can be computed in polynomial time given value-oracle access.

Proof This is one iteration of the algorithm of Barman et al. [BKNS22]. If (A, p) is extendable with g , that is, if some permutation of the bundles keeps the solution envy-free and gives a maximally subsidised agent a bundle on which g has marginal value one, the subroutine **EXTEND** finds such a permutation and agent, and that agent receives g (Lemmas 7 and 8 of [BKNS22]). Otherwise the subroutine **FINDSINK** finds a recipient for g (Lemmas 9 to 11 of [BKNS22]). In both cases the pointwise-minimal subsidy vector of the new allocation lies in $\{0, 1\}^n$, and the step runs in polynomial time.

These lemmas are stated for instances in which every item is a good, whereas the bundles of A may contain chores. Their proofs, however, use that hypothesis only through two facts: the valuations are integer valued, and the good being inserted has marginal value in $\{0, 1\}$ for every agent on every bundle. Monotonicity and dichotomy are invoked only for that good, never for the items already allocated. Both facts hold here, so the step applies unchanged. Appendix C restates the argument and checks each step under these two facts. $\square$

We can now obtain the unit-subsidy guarantee for objective signed dichotomous valuations.

Theorem 26. *For every instance with objective signed dichotomous valuations, there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and*

$$\sum_{i \in N} p_i \leq n - 1.$$

Moreover, such an allocation and subsidy vector can be computed in polynomial time in the value-oracle model.

Proof Restrict the instance to the chores C . By Theorem 22, we can compute an envy-free solution $(\mathcal{A}^C, p^C)$ with $p^C \in \{0, 1\}^n$. Starting from this solution, insert the goods in G one at a time using Lemma 25. After all insertions, we obtain a complete envy-free solution $(\mathcal{A}, p)$ with $p \in \{0, 1\}^n$. If every component of p equals one, subtract one from every subsidy. Otherwise some component is already zero. Thus, after normalization, $\sum_i p_i \leq n - 1$. Polynomial-time computability follows from Theorem 22 and Lemma 25. $\square$

The resulting procedure is stated explicitly as Algorithm 2. For brevity, let **CHORE-ALLOCATE** denote the polynomial-time procedure of Section 3 (Algorithm 1) for negative dichotomous valuations. Given a negative dichotomous chore instance, it returns a complete allocation $\mathcal{A}$ and $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and $\sum_i p_i \leq n - 1$.

Algorithm 2 Unit subsidies for objective signed dichotomous valuations

Require: Agents $N = [n]$, items $M = G \cup C$, and value-oracle access to objective signed dichotomous valuations $\{v_i\}_{i \in N}$.

Ensure: A complete allocation $\mathcal{A}$ and $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and $\sum_{i \in N} p_i \leq n - 1$.

- 1: Restrict each v_i to subsets of C .
 - 2: $(\mathcal{A}, p) \leftarrow \text{CHOREALLOCATE}(N, C, \{v_i|_C\}_{i \in N})$.
 - 3: Replace p by the pointwise-minimal subsidy vector supporting $\mathcal{A}$.
 - 4: **for** each $g \in G$ **do**
 - 5: Apply the one-good extension procedure of Lemma 25 to $(\mathcal{A}, p)$ and g , obtaining an allocation $\mathcal{A}'$ of the currently allocated items together with g and a subsidy vector $p' \in \{0, 1\}^n$ such that $(\mathcal{A}', p')$ is envy-free.
 - 6: $\mathcal{A} \leftarrow \mathcal{A}'$.
 - 7: Replace p' by the pointwise-minimal subsidy vector supporting $\mathcal{A}$, and set $p \leftarrow p'$.
 - 8: **return** $(\mathcal{A}, p)$.
-

The one-good extension in Line 5 is implemented by the `EXTEND/FINDSINK` procedure of Barman et al. [BKNS22], which remains valid in the present setting by Lemma 25. After every iteration, we take the pointwise-minimal supporting subsidy vector. Since the valuations are integer-valued, this vector is integral, and Lemma 25 implies that all its components are at most one. Moreover, pointwise minimality implies that at least one component is zero. Hence, throughout the algorithm

$$p \in \{0, 1\}^n \quad \text{and} \quad \sum_{i \in N} p_i \leq n - 1.$$

The total-subsidy bound in Theorem 26 is tight. Indeed, negative dichotomous chore instances form a subclass of objective signed dichotomous valuations, and the identical binary additive instance with n agents and $n - 1$ chores of Example 1 already requires a total subsidy of at least $n - 1$.

Theorem 26 concerns the subsidy guarantee only. Our construction does not guarantee that the underlying allocation is EF1, since the subsequent goods-allocation procedure of Barman et al. [BKNS22] does not in general preserve such a guarantee. Pareto optimality also cannot be imposed in general, since Proposition 28 applies to the pure-chore subclass. Under additivity, however, Pareto optimality can be recovered together with the same subsidy bound (Proposition 29).

5 Pareto Optimality and Unit Subsidies

We next examine whether the unit-subsidy guarantee can be obtained together with Pareto optimality of the underlying allocation. Pareto optimality is evaluated with respect to the allocation of items, without taking subsidies into account. Formally, a complete allocation $\mathcal{B}$ *Pareto dominates* a complete allocation $\mathcal{A}$ if

$$v_i(\mathcal{B}_i) \geq v_i(\mathcal{A}_i) \quad \text{for every } i \in N,$$

with strict inequality for at least one agent. In cost form this reads $c_i(\mathcal{B}_i) \leq c_i(A_i)$. A complete allocation is *Pareto optimal* if it is not Pareto dominated by any other complete allocation. The quantifier ranges over complete allocations: for chores, withholding an item weakly benefits every agent, so allowing incomplete allocations would leave no allocation Pareto optimal.

5.1 Negative dichotomous valuations

For the binary additive subclass, Pareto optimality is compatible with the unit-subsidy guarantee. This parallels the corresponding phenomenon for binary additive goods, where Halpern and Shah [HS19] obtain a Pareto-efficient allocation requiring total subsidy at most $n - 1$. See also Goko et al. [GIK⁺24] for unit subsidies together with utilitarian optimality in the more general matroidal goods setting. For completeness, we record the simple chore-side observation.

Proposition 27. *For binary additive costs, there exists a complete allocation $\mathcal{A}$ and $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free, $\sum_{i \in N} p_i \leq n - 1$, and $\mathcal{A}$ is Pareto optimal and EF1. Moreover, such an outcome can be computed in polynomial time.*

Proof Let

$$U := \{e \in M : c_i(\{e\}) = 1 \text{ for every } i \in N\}$$

be the universally costly chores. Assign every $e \notin U$ to an agent for whom $c_i(e) = 0$. Write $|U| = qn + r$, where $0 \leq r < n$, and distribute the chores in U so that r agents receive $q + 1$ of them and the remaining agents receive q .

Set $p_i = 1$ for the former r agents and $p_i = 0$ for the others. Then $c_i(A_i) - p_i = q$ for every i . Moreover, every bundle A_j contains $q + p_j$ universally costly chores, and hence

$$c_i(A_j) - p_j \geq q = c_i(A_i) - p_i$$

for all i, j . Thus $(\mathcal{A}, p)$ is envy-free and $\sum_i p_i = r \leq n - 1$.

Every chore outside U is allocated at zero cost, whereas every chore in U necessarily contributes one unit of cost under every allocation. Hence $\mathcal{A}$ minimizes total cost and is therefore Pareto optimal. Finally, an agent receiving q universally costly chores is already envy-free, while an agent receiving $q + 1$ becomes envy-free after removing one universally costly chore. Thus $\mathcal{A}$ is EF1. The construction clearly requires only polynomially many value queries and operations. $\square$

Thus, in the binary additive domain, Pareto optimality is compatible with the optimal unit-subsidy guarantee. The total-subsidy bound remains tight even under Pareto optimality: in the identical binary additive instance $c_i(S) = |S|$ of Example 1, every complete allocation is Pareto optimal, while every envy-free solution requires total subsidy at least $n - 1$. This compatibility, however, breaks down once nonadditivity is allowed, even for identical binary submodular costs.

Proposition 28. *There exist negative dichotomous instances, even with two agents and identical binary submodular cost functions, for which no Pareto-optimal allocation admits an envy-free subsidy vector in $\{0, 1\}^n$.*

Proof Consider two agents $N = \{1, 2\}$ and three chores $M = \{a, b, c\}$. Both agents have the same cost function

$$c_1(S) = c_2(S) = \min\{|S|, 2\}.$$

Every marginal cost belongs to $\{0, 1\}$, so the cost function is dichotomous. Moreover, it is submodular, since $k \mapsto \min\{k, 2\}$ is a nondecreasing concave function of cardinality. It is not additive, since

$$c_i(\{a, b, c\}) = 2 < 3 = \sum_{g \in \{a, b, c\}} c_i(\{g\}).$$

Every complete allocation has, up to symmetry, either bundle-size profile $(3, 0)$ or $(2, 1)$. First, consider the allocation in which agent 1 receives all three chores. Its cost profile is $(2, 0)$. The possible cost profiles of complete allocations are

$$(2, 0), \quad (2, 1), \quad (1, 2), \quad (0, 2),$$

and none of the latter three Pareto dominates $(2, 0)$. Hence, the concentrated allocation is Pareto optimal. By symmetry, the allocation in which agent 2 receives all three chores is also Pareto optimal. However, a concentrated allocation cannot be made envy-free with unit subsidies. If agent 1 receives all three chores, envy-freeness requires

$$2 - p_1 \leq -p_2,$$

and hence

$$p_1 - p_2 \geq 2.$$

Thus, no subsidy vector in $\{0, 1\}^2$ suffices. The case in which agent 2 receives all three chores is symmetric.

Now consider a split allocation, with bundle sizes $(2, 1)$ up to symmetry. Its cost profile is $(2, 1)$, so it can be made envy-free with subsidies $(1, 0)$, up to symmetry. Nevertheless, it is not Pareto optimal. Transferring the singleton chore to the agent holding two chores changes the cost profile from $(2, 1)$ to $(2, 0)$: the cost of one agent remains unchanged, while the other agent's cost strictly decreases.

Thus, the Pareto-optimal allocations are precisely the two concentrated allocations, and each requires a subsidy of at least 2 for the agent receiving all three chores. Conversely, every allocation that admits an envy-free subsidy vector in $\{0, 1\}^2$ is Pareto dominated. Therefore, no allocation is simultaneously Pareto optimal and envy-free with subsidies in $\{0, 1\}^2$. $\square$

All eight complete allocations of this instance, with their minimal subsidies, are listed in Example 40 of Appendix B. Thus, the unit-subsidy guarantee of Theorem 22 cannot, in general, be strengthened to require Pareto optimality.

5.2 Additive objective signed dichotomous valuations

For objective signed dichotomous valuations (Definition 24), Proposition 28 already rules out Pareto optimality in general, since negative dichotomous valuations form a subclass. Under additivity, however, Pareto optimality can be recovered together with the subsidy bound of Theorem 26.

Proposition 29. *For every instance with additive objective signed dichotomous valuations, there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free,*

$$\sum_{i \in N} p_i \leq n - 1,$$

and $\mathcal{A}$ is Pareto optimal. Moreover, such an outcome can be computed in polynomial time.

Proof The proof has three steps. First, allocate the goods using Halpern and Shah’s binary-additive result, obtaining an envy-freeable allocation with unit subsidies. Next, assign every chore that is costless to some agent to such an agent. Finally, assign each universally costly chore to an agent currently receiving zero subsidy and raise that agent’s subsidy from 0 to 1. If all subsidies become 1, subtract one from every subsidy. This preserves envy-freeness and keeps total subsidy at most $n - 1$. Since every item is assigned to an agent attaining its maximum possible value, the final allocation is Pareto optimal.

We now formally write the details. Let $M = G \dot{\cup} C$ be the common partition into goods and chores. Since the valuations are additive, for every agent i ,

$$v_i(g) \in \{0, 1\} \quad (g \in G)$$

and

$$v_i(c) \in \{0, -1\} \quad (c \in C).$$

We first allocate the goods. Ignore, for the moment, goods that have value zero for every agent, and apply the binary-additive goods result of Halpern and Shah [HS19] to the remaining goods. Their construction returns a non-wasteful allocation $\mathcal{A}^G$ that is envy-freeable and whose minimal subsidies have total at most $n - 1$. Moreover, their proof shows that every path in the corresponding weighted envy graph has weight at most one. Since all edge weights are integral, the pointwise-minimal subsidies are integral as well, and hence we may choose

$$p \in \{0, 1\}^n, \quad \sum_{i \in N} p_i \leq n - 1.$$

Goods having value zero for every agent may now be allocated arbitrarily, without affecting either envy-freeness or the subsidy vector.

We next allocate the chores. Partition C into

$$C^0 := \{c \in C : \text{there exists } i \in N \text{ with } v_i(c) = 0\}$$

and

$$C^- := \{c \in C : v_i(c) = -1 \text{ for every } i \in N\}.$$

Thus, C^- is the set of universally costly chores.

First, consider a chore $c \in C^0$. Choose an agent k with $v_k(c) = 0$ and assign c to k . This preserves envy-freeness with the same subsidy vector. Indeed, agent k ’s value for her own bundle is unchanged, whereas for every agent $i \neq k$,

$$v_i(A_k \cup \{c\}) = v_i(A_k) + v_i(c) \leq v_i(A_k).$$

Hence, k 's bundle becomes weakly less desirable to every other agent, while all other relevant bundle values remain unchanged. Repeating this operation allocates all chores in C^0 while preserving the envy-free solution $(\mathcal{A}, p)$.

It remains to allocate the universally costly chores in C^- . We maintain the following invariant:

$$(\mathcal{A}, p) \text{ is envy-free, } p \in \{0, 1\}^n, \text{ and } p_k = 0 \text{ for some } k \in N. \quad (7)$$

The invariant holds initially. In particular, the pointwise-minimal subsidy vector has at least one zero component.

Let $c \in C^-$ be an unallocated chore, and choose an agent k with $p_k = 0$. Assign c to k and define

$$\widehat{p}_k := 1, \quad \widehat{p}_i := p_i \quad (i \neq k).$$

We claim that the resulting solution remains envy-free.

Since c has value -1 for every agent,

$$v_k(A_k \cup \{c\}) + \widehat{p}_k = v_k(A_k) - 1 + 1 = v_k(A_k) + p_k.$$

Thus, agent k 's subsidized value for her own bundle is unchanged. Moreover, for every $i \neq k$,

$$v_i(A_k \cup \{c\}) + \widehat{p}_k = v_i(A_k) - 1 + 1 = v_i(A_k) + p_k.$$

Hence, the subsidized value of k 's bundle is unchanged from the viewpoint of every agent. All other bundles and subsidies are unchanged. Consequently, every envy-freeness inequality is exactly the same as before, and the new solution is envy-free.

If $\widehat{p}$ has a zero component, set $p := \widehat{p}$ and continue. Otherwise, if $\widehat{p}_i = 1$ for all $i \in N$, replace it with

$$p_i := \widehat{p}_i - 1 = 0 \text{ for all } i \in N.$$

Subtracting the same amount from every agent's subsidy leaves all envy-freeness inequalities unchanged. Thus, the invariant (7) is restored. Repeating this procedure allocates every chore in C^- and produces a complete allocation $\mathcal{A}$ with

$$p \in \{0, 1\}^n \quad \text{and} \quad \sum_{i \in N} p_i \leq n - 1.$$

It remains to prove Pareto optimality. By construction, every good that is valued at one by some agent is assigned to an agent who values it at one. A good valued at zero by every agent may be assigned arbitrarily. Every chore in C^0 is assigned to an agent who values it at zero, while every chore in C^- has value -1 for every agent regardless of its recipient. Therefore, every item is assigned to an agent attaining the maximum possible value for that item. By additivity,

$$\sum_{i \in N} v_i(A_i) = \sum_{e \in M} \max_{i \in N} v_i(e),$$

which is the maximum possible utilitarian welfare over all complete allocations. Hence $\mathcal{A}$ is Pareto optimal.

Finally, the binary-additive goods allocation and its supporting subsidies can be computed in polynomial time by the construction of Halpern and Shah [HS19]. The subsequent chore-allocation steps require only identifying a zero-valued recipient for each chore and updating a binary subsidy vector. Hence, the entire construction runs in polynomial time. $\square$

6 Goods First, and a Reduction for Doubly Monotone Valuations

Theorem 26 allocates the chores first and inserts the goods afterwards. That order has no counterpart for doubly monotone valuations, in which an item may be a good for some agents and a chore for others: such an item cannot be set aside as a chore before it has been tried as a good. This section reverses the order. We show that items that are chores for every agent can be added at the end to any envy-free solution with unit subsidies, at the cost of at most one more unit per agent and $n - 1$ more in total (Lemma 34). For objective signed dichotomous valuations this gives a second, goods-first algorithm with at most two units per agent (Theorem 35). For doubly monotone dichotomous valuations it reduces the same guarantee to a single insertion property for the remaining items (Proposition 38).

Definition 30 (Doubly monotone dichotomous valuations, universal chores). *A valuation profile $\{v_i\}_{i \in N}$ is doubly monotone dichotomous if for every agent i there is a partition $M = G_i \dot{\cup} C_i$ such that, for every $S \subseteq M$,*

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, 1\} \quad \text{for every } g \in G_i \setminus S,$$

and

$$v_i(S \cup \{c\}) - v_i(S) \in \{0, -1\} \quad \text{for every } c \in C_i \setminus S.$$

An item e is a universal chore if $v_i(S \cup \{e\}) - v_i(S) \in \{0, -1\}$ for every agent i and every $S \subseteq M \setminus \{e\}$.

Objective signed dichotomous valuations (Definition 24) are the case $G_i = G$ and $C_i = C$ for every i , and there every item of C is a universal chore. Doubly monotone dichotomous valuations are integer valued, by the telescoping argument of Observation 11. Hence every envy graph has integer arc weights, every pointwise-minimal subsidy vector is integral, and Theorems 9 and 10 and Proposition 12 apply to allocations of any set of items, since their proofs use only the arc weights.

6.1 Packages and the tie graph

Fix a set $M' \subseteq M$, an allocation $A = (A_1, \dots, A_n)$ of M' and an integral subsidy vector p . Call (A_j, p_j) the *package* of agent j , and write

$$U_i(j) := v_i(A_j) + p_j \tag{8}$$

for its value to agent i . Then (A, p) is an envy-free solution if and only if $U_i(i) \geq U_i(j)$ for all $i, j \in N$. All values $U_i(j)$ are integers, so in an envy-free solution $U_i(j) \neq U_i(i)$ implies $U_i(j) \leq U_i(i) - 1$. The *tie graph* of (A, p) is

$$H = (N, E), \quad (i, j) \in E \iff i \neq j \text{ and } U_i(i) = U_i(j). \tag{9}$$

It is the equality graph (3) of Section 3.1 with every bundle replaced by its package. The payments must enter it. If agent i prefers A_i to the bundle of a paid agent j by exactly one, then i is indifferent between the two packages, and after one more costly chore i would envy the package of j . In the tie graph this indifference is the arc (i, j) .

6.2 Allocating the universal chores

Let (A, p) be an envy-free solution for an allocation A of $M' \subseteq M$ with $p \in \{0, 1\}^n$, and suppose that every item of $R := M \setminus M'$ is a universal chore. Starting from (A, p) , apply the following rules in the stated order while $R \neq \emptyset$, recomputing H after every step and removing from R every chore that is assigned. The procedure halts when $R = \emptyset$, or at the final clause of (U3).

- (U1) If some $e \in R$ and some $i \in N$ satisfy $v_i(A_i \cup \{e\}) = v_i(A_i)$, assign e to i .
- (U2) If some $e \in R$ and some arc $(i, j) \in E$ lying on a directed cycle of H satisfy $v_i(A_j \cup \{e\}) = v_i(A_j)$, rotate the packages along that cycle, so that every agent on the cycle receives the bundle and the subsidy of the next agent on the cycle, and then assign e to i .
- (U3) Otherwise select a strongly connected component S of H with no arc of E leaving S . If $|R| \geq |S|$, assign one chore of R to each agent of S , distinct agents receiving distinct chores. Otherwise halt.

These are the rules (R1) to (R3) of Section 3.1, with the equality graph replaced by the tie graph and with rotations that move each subsidy together with its bundle.

Lemma 31. *Each application of (U1), (U2) or (U3) assigns at least one chore of R and turns (A, p) into an envy-free solution whose subsidy vector is a permutation of the previous one. Consequently the procedure halts after at most $|M \setminus M'|$ applications, and throughout it the subsidy vector lies in $\{0, 1\}^n$ and keeps its sum.*

Proof Every rule assigns at least one chore, (U3) assigning $|S| \geq 1$ of them, and only (U2) moves subsidies, by a cyclic permutation. It remains to show that each step preserves envy-freeness. Write U' for the package values after the step. Throughout we use that adding a universal chore to a bundle never raises any agent's value for it.

(U1). The package of i keeps its value to i , because e is free for i on A_i , and it is worth at most as much as before to every other agent. All other packages are unchanged. Hence $U'_k(k) = U_k(k) \geq U_k(j) \geq U'_k(j)$ for all $k, j \in N$.

(U2). Let the cycle be $i = a_0 \rightarrow a_1 \rightarrow \dots \rightarrow a_{L-1} \rightarrow a_0$, with $a_1 = j$. After the step, an agent a_t with $t \neq 0$ holds the package $(A_{a_{t+1}}, p_{a_{t+1}})$, indices taken modulo L , worth $U_{a_t}(a_{t+1}) = U_{a_t}(a_t)$ to a_t because $(a_t, a_{t+1}) \in E$. Agent i holds $(A_j \cup \{e\}, p_j)$, worth $v_i(A_j \cup \{e\}) + p_j = v_i(A_j) + p_j = U_i(j) = U_i(i)$ to i , because e is free for i on A_j and $(i, j) \in E$. Agents off the cycle keep their packages. So every agent's own package keeps its value. The packages after the step are those before it, except that (A_j, p_j) has become $(A_j \cup \{e\}, p_j)$, which no agent values more. Hence every agent k values every package after the step at most $\max_{j \in N} U_k(j) = U_k(k) = U'_k(k)$.

(U3). Neither (U1) nor (U2) applies. Agents outside S keep their packages, and no package gains value to anyone, so every $k \notin S$ satisfies $U'_k(k) = U_k(k) \geq U_k(j) \geq U'_k(j)$ for every j . Let

$i \in S$ and let e_i be the chore assigned to i . As (U1) does not apply, e_i costs i one unit on A_i , so $U'_i(i) = U_i(i) - 1$. Let $j \neq i$.

- If $j \notin S$, then $(i, j) \notin E$ because no arc leaves S . So $U_i(j) \leq U_i(i) - 1$, and $U'_i(j) = U_i(j) \leq U'_i(i)$.
- If $j \in S$ and $(i, j) \in E$, then a path from j to i inside S closes a directed cycle through (i, j) , since S is strongly connected. As (U2) does not apply, the chore e_j assigned to j costs i one unit on A_j . So $U'_i(j) = U_i(j) - 1 = U_i(i) - 1 = U'_i(i)$.
- If $j \in S$ and $(i, j) \notin E$, then $U_i(j) \leq U_i(i) - 1$, and $U'_i(j) \leq U_i(j) \leq U'_i(i)$.

Thus the solution after the step is envy-free. $\square$

Lemma 32. *Suppose the procedure halts with $R \neq \emptyset$. Let (A, p) be the solution at which it halts, with tie graph $H = (N, E)$ and package values U , let $r := |R|$, and let S be the component selected by (U3) at the halting step. Then the following hold.*

- (a) $r < |S|$.
- (b) $v_i(A_i \cup \{e\}) = v_i(A_i) - 1$ for every $i \in N$ and every $e \in R$.
- (c) $v_i(A_j \cup \{e\}) = v_i(A_j) - 1$ for every arc $(i, j) \in E$ with $i, j \in S$ and every $e \in R$.
- (d) $U_i(j) \leq U_i(i) - 1$ for every $i \in S$ and every $j \in N \setminus S$.

Proof The procedure halts at the final clause of (U3), so neither (U1) nor (U2) applies and $|R| < |S|$, which is (a). Every chore of R is universal, so its marginals lie in $\{0, -1\}$. A marginal 0 in (b) would let (U1) apply. In (c), a path from j to i inside S closes a directed cycle through (i, j) , so a marginal 0 would let (U2) apply. For (d), no arc leaves S , so $U_i(j) \neq U_i(i)$, and envy-freeness gives $U_i(j) \leq U_i(i) - 1$. $\square$

6.3 The completion

Suppose the procedure halts with $R = \{e_1, \dots, e_r\} \neq \emptyset$, and keep the notation of Lemma 32. By its part (a) we may fix r distinct agents $T = \{t_1, \dots, t_r\} \subseteq S$, chosen arbitrarily. Define the complete allocation A^* by

$$A_{t_k}^* := A_{t_k} \cup \{e_k\} \quad (1 \leq k \leq r), \quad A_i^* := A_i \quad (i \notin T).$$

Let $P \subseteq N$ be the smallest set such that

- (P1) $T \subseteq P$, and
- (P2) if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$,

and set

$$m_i := p_i + \mathbf{1}[i \in P] \quad (i \in N). \quad (10)$$

These are the recipients and the subsidy set of Section 3.2, built on the tie graph. Every agent added to P by (P2) lies outside S , so $P \setminus T \subseteq N \setminus S$, exactly as in Lemma 17.

Lemma 33. *Let $i \in S$, let $t \in T$ and let $e \in R$. Then $v_i(A_t \cup \{e\}) + p_t \leq U_i(i) - 1$.*

Proof Envy-freeness gives $U_i(t) \leq U_i(i)$. If $U_i(t) < U_i(i)$, then $U_i(t) \leq U_i(i) - 1$, and $v_i(A_t \cup \{e\}) \leq v_i(A_t)$ because e is a universal chore. Otherwise $U_i(t) = U_i(i)$. If $i = t$, part (b) of Lemma 32 gives $v_i(A_i \cup \{e\}) + p_i = U_i(i) - 1$. If $i \neq t$, then $(i, t) \in E$ with $i, t \in S$, and part (c) gives $v_i(A_t \cup \{e\}) + p_t = U_i(t) - 1 = U_i(i) - 1$. $\square$

Lemma 34 (Universal chores at one more unit per agent). *Let $\{v_i\}_{i \in N}$ be doubly monotone dichotomous, let A be an allocation of a set $M' \subseteq M$, and let $p \in \{0, 1\}^n$ be such that (A, p) is an envy-free solution. Suppose that every item of $M \setminus M'$ is a universal chore. Then there are a complete allocation A^* and a subsidy vector $m \in \{0, 1, 2\}^n$ such that (A^*, m) is an envy-free solution and*

$$\sum_{i \in N} m_i \leq \sum_{i \in N} p_i + n - 1.$$

More precisely, m exceeds a permutation of p by a vector in $\{0, 1\}^n$. Moreover, (A^, m) can be computed in polynomial time given value-oracle access.*

Proof Run the procedure of Section 6.2. By Lemma 31 it halts at an envy-free solution whose subsidy vector, again written p , is a permutation of the original one. If it halts with $R = \emptyset$, return this solution with $m = p$. Otherwise construct A^* and m as in Section 6.3. The allocation A^* is complete, since the t_k are distinct and every chore of R is used once. The vector $m - p$ is the indicator of P , so $m \in \{0, 1, 2\}^n$. Since $P \subseteq T \cup (N \setminus S)$, $|T| = r$ and $r \leq |S| - 1$ by Lemma 32(a),

$$\sum_{i \in N} m_i = \sum_{i \in N} p_i + |P| \leq \sum_{i \in N} p_i + r + n - |S| \leq \sum_{i \in N} p_i + n - 1.$$

It remains to show that (A^*, m) is envy-free. Write $U_i^*(j) := v_i(A_j^*) + m_j$. Since $A_j \subseteq A_j^*$ and the item added to A_j , if any, is a universal chore,

$$U_i^*(j) \leq U_i(j) + \mathbf{1}[j \in P], \quad \text{with equality when } j \notin T.$$

Fix $i, j \in N$. Since $T \subseteq P$, the following three cases are exhaustive.

Case 1: $i \in T$. Then $i \in S$ and $i \in P$, and Lemma 32(b) gives $U_i^*(i) = U_i(i) - 1 + 1 = U_i(i)$.

(1a) $j \in T$. Writing e for the chore assigned to j , Lemma 33 gives $U_i^*(j) = v_i(A_j \cup \{e\}) + p_j + 1 \leq U_i(i)$.

(1b) $j \in P \setminus T$. Then $j \notin S$, so Lemma 32(d) gives $U_i^*(j) = U_i(j) + 1 \leq U_i(i)$.

(1c) $j \notin P$. Then $U_i^*(j) = U_i(j) \leq U_i(i)$.

Case 2: $i \in P \setminus T$. Then $A_i^* = A_i$, so $U_i^*(i) = U_i(i) + 1 \geq U_i(j) + 1 \geq U_i^*(j)$ for every j .

Case 3: $i \notin P$. Then $i \notin T$, so $U_i^*(i) = U_i(i)$.

(3a) $j \notin P$. Then $U_i^*(j) = U_i(j) \leq U_i(i)$.

(3b) $j \in P$ and $i \notin S$. If $U_i(j) = U_i(i)$, then $i \neq j$ because $j \in P$ and $i \notin P$, so $(i, j) \in E$ and (P2) puts i in P , a contradiction. Hence $U_i(j) \leq U_i(i) - 1$ and $U_i^*(j) \leq U_i(j) + 1 \leq U_i(i)$.

(3c) $j \in P \setminus T$ and $i \in S$. Then $j \notin S$, so Lemma 32(d) gives $U_i^*(j) = U_i(j) + 1 \leq U_i(i)$.

(3d) $j \in T$ and $i \in S$. Writing e for the chore assigned to j , Lemma 33 gives $U_i^*(j) = v_i(A_j \cup \{e\}) + p_j + 1 \leq U_i(i)$.

In every case $U_i^*(i) \geq U_i^*(j)$.

For the running time, the tie graph is determined by the n^2 values $v_i(A_j)$, its strongly connected components and the cycles used by (U2) are found by graph search, each rule tests at most $n^2 |R|$ marginals, and by Lemma 31 there are at most $|M \setminus M'|$ rule applications. Computing P is a backward reachability search in H . $\square$

6.4 A goods-first algorithm for objective signed valuations

Theorem 35. *For every instance with objective signed dichotomous valuations, the following procedure computes, in polynomial time given value-oracle access, a complete allocation $\mathcal{A}$ and a subsidy vector $m \in \{0, 1, 2\}^n$ such that $(\mathcal{A}, m)$ is envy-free and $\sum_{i \in N} m_i \leq 2(n - 1)$. Allocate the goods G by the algorithm of Barman et al. [BKNS22], replace its subsidy vector by the pointwise-minimal one, and allocate the chores C by Lemma 34.*

Proof On subsets of G every v_i has marginals in $\{0, 1\}$, so the algorithm of [BKNS22] returns an allocation A of G and $p \in \{0, 1\}^n$ with (A, p) envy-free. By Theorem 10 the pointwise-minimal subsidy vector is at most p componentwise. It is integral, so it lies in $\{0, 1\}^n$, and by Proposition 12 it has a zero coordinate, so its sum is at most $n - 1$. Every chore of C is a universal chore, so Lemma 34 applies with $M' = G$ and gives $m \in \{0, 1, 2\}^n$ with $\sum_i m_i \leq (n - 1) + (n - 1)$. Both steps run in polynomial time. $\square$

Since the pointwise-minimal subsidy vector of $\mathcal{A}$ is at most m componentwise, it satisfies the same two bounds. Theorem 35 is weaker than Theorem 26, which pays at most one unit per agent. Its interest is the order of allocation, which is the order that Section 6.5 needs.

Remark 36 (A chore with one unit of money attached). A tempting alternative completion attaches one unit of subsidy to each residual chore and inserts the pair as a single item. The marginal value of the pair formed from a chore e is $v_i(S \cup \{e\}) - v_i(S) + 1 \in \{0, 1\}$ for every agent and every bundle, so the pair behaves as a dichotomous good. Formally, replace each residual chore e by a new item $\widehat{e}$, let $\widehat{R}$ be the set of new items, and apply Lemma 25 to the valuations

$$V_i(S) := v_i((S \setminus \widehat{R}) \cup \{e : \widehat{e} \in S\}) + |S \cap \widehat{R}|.$$

This variant keeps the total bound. At most $n - 1$ pairs are formed, since the procedure of Section 6.2 halts with $r < |S| \leq n$, and Lemma 25 keeps the remaining subsidy vector in $\{0, 1\}^n$, with a zero coordinate after passing to the pointwise-minimal vector. So the payments total at most $2(n - 1)$. It loses the per-agent bound, because the units attached to the pairs accumulate when one agent receives several of them, as Example 37 shows. The distinct recipients of Section 6.3 prevent exactly this.

Example 37. Take four agents x, a, b, d , goods g_1, g_2, g_3 and chores c_1, c_2 , and give every agent the additive valuation $v(S) = |S \cap G| - |S \cap C|$. In the algorithm of [BKNS22] every good is extendable, so it goes to a maximally subsidized agent with marginal value one, and

the maximally subsidized agents are exactly those holding no good. So, up to renaming, the goods phase returns $A_x = \emptyset$, $A_a = \{g_1\}$, $A_b = \{g_2\}$ and $A_d = \{g_3\}$, with subsidy 1 for x and 0 for the others. Every agent values every package at 1, so the tie graph is complete and $S = N$. Since $|R| = 2 < 4$ and every chore costs every agent one unit on every bundle, the procedure of Section 6.2 halts at once.

Each pair $\widehat{c}_k$ has marginal value 0 for every agent on every bundle, so it is not extendable, and FINDSINK starts at x , the only maximally subsidized agent. Giving $\widehat{c}_1$ to x changes no agent's value for any package, so the minimal subsidies stay $(1, 0, 0, 0)$, no agent reaches 2, and FINDSINK returns x . The same happens for $\widehat{c}_2$. Agent x ends with $\{c_1, c_2\}$, two attached units and a subsidy of 1, so x is paid 3. The final allocation forces this, since $v_x(A_a) - v_x(\{c_1, c_2\}) = 3$, so by Theorem 10 no envy-free subsidy vector pays x less than 3. With n agents, $n - 1$ such goods and $n - 2$ such chores, the same run pays x exactly $n - 1$.

The completion of Section 6.3 instead gives c_1 and c_2 to two distinct agents of S . For $T = \{a, b\}$ it pays x , a and b one unit each, and every agent ends with value plus subsidy equal to 1.

6.5 Doubly monotone valuations

Lemma 34 needs nothing from the items allocated before it beyond an envy-free solution with unit subsidies. This reduces the two-unit guarantee for doubly monotone dichotomous valuations to the following insertion property.

(INS) For every set $M' \subseteq M$, every allocation A of M' and every $p \in \{0, 1\}^n$ such that (A, p) is envy-free, and every item $e \in M \setminus M'$ that is not a universal chore, there exist an allocation A' of $M' \cup \{e\}$ and $p' \in \{0, 1\}^n$ such that (A', p') is envy-free.

Proposition 38. *Let $\{v_i\}_{i \in N}$ be doubly monotone dichotomous and satisfy (INS). Then there are a complete allocation $\mathcal{A}$ and a subsidy vector $m \in \{0, 1, 2\}^n$ such that $(\mathcal{A}, m)$ is envy-free and $\sum_{i \in N} m_i \leq 2(n - 1)$.*

Proof Start from the empty allocation with zero subsidies, which is envy-free. Insert the items that are not universal chores one at a time by (INS), each time replacing the subsidy vector by the pointwise-minimal one. That vector is integral and componentwise at most the vector given by (INS), so it lies in $\{0, 1\}^n$. When these items are exhausted, the current solution is envy-free, its subsidy vector lies in $\{0, 1\}^n$ and has sum at most $n - 1$ by Proposition 12, and every unallocated item is a universal chore. Lemma 34 completes it with $m \in \{0, 1, 2\}^n$ and $\sum_i m_i \leq (n - 1) + (n - 1)$. $\square$

Lemma 25 proves (INS) for every item that is a good for every agent, that is, whose marginal value lies in $\{0, 1\}$ for every agent and every bundle, whatever the current bundles contain. For objective signed valuations every item that is not a universal chore is of this kind, so Proposition 38 also yields the bound of Theorem 35. The open case is an item that is a good for some agents and a chore for others. The argument behind Lemma 25 uses at several points that the new item has a nonnegative marginal value for the agent receiving it, for instance in Claim 2 and in the inequality $\text{SW}(X^{s_0}) \geq \text{SW}(A)$ of Case 2 in Appendix C, and this fails when such an item is tentatively given to an agent for whom it is a chore. Whether (INS) holds for such items is open.

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A Failure of a BKNS-type incremental construction

Example 39 (Failure of a BKNS-type incremental extension [BKNS22]). Consider three agents $N = \{1, 2, 3\}$ and three chores $M = \{a, b, c\}$. We work in cost form, writing $c_i = -v_i$ for agent i 's non-negative cost function. Thus an allocation $\mathcal{A} = (A_1, \dots, A_n)$ together with a subsidy vector $p = (p_1, \dots, p_n)$ is envy-free if and only if

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N.$$

The cost functions depend only on bundle size:

$$c_1(S) = \max\{0, |S| - 1\}, \quad c_2(S) = c_3(S) = \min\{|S|, 2\}.$$

Thus

$ S $	0	1	2	3
$c_1(S)$	0	0	1	2
$c_2(S) = c_3(S)$	0	1	2	2

Every marginal cost belongs to $\{0, 1\}$, so these are dichotomous cost functions.

Consider the following two steps of an incremental construction.

step	allocation	p^*	$M(p^*)$
start	$(\emptyset, \emptyset, \emptyset)$	$(0, 0, 0)$	$\{1, 2, 3\}$
a to agent 1	$(\{a\}, \emptyset, \emptyset)$	$(0, 0, 0)$	$\{1, 2, 3\}$
b to agent 1	$(\{a, b\}, \emptyset, \emptyset)$	$(1, 0, 0)$	$\{1\}$

Both steps are legal: at each one the recipient lies in $M(p^*)$ and the resulting allocation is envy-freeable with subsidies in $\{0, 1\}^3$.

Now c is currently unallocated and the partial allocation is

$$X_1 = \{a, b\}, \quad X_2 = X_3 = \emptyset,$$

whose pointwise-minimal subsidy vector is $p^* = (1, 0, 0)$. Indeed, agent 1 has cost 1 for her own bundle and cost 0 for either empty bundle, so she requires one unit of subsidy; agents 2 and 3 have no envy, since they have cost 0 for their own empty bundles and cost 2 for X_1 .

We claim that the remaining chore c cannot be inserted so that the resulting allocation admits an envy-free subsidy vector in $\{0, 1\}^3$, even if the existing bundles $\{a, b\}, \emptyset, \emptyset$ may first be permuted among the agents.

If c is added to the two-chore bundle, the resulting bundle sizes are $(3, 0, 0)$. Every agent evaluates a three-chore bundle at cost 2 and an empty bundle at cost 0. Hence the agent receiving the three-chore bundle requires a subsidy at least two units larger than that of an agent receiving an empty bundle, and no subsidy vector in $\{0, 1\}^3$ can make the resulting allocation envy-free.

It remains to consider the case in which c is added to an empty bundle. The resulting bundle sizes are $(2, 1, 0)$, up to permutation. If an agent in $\{2, 3\}$ receives the two-chore bundle, she evaluates it at cost 2 and the empty bundle at cost 0, again requiring a subsidy difference of at least two.

Therefore the only remaining possibility is that agent 1 receives the two-chore bundle. Let $i \in \{2, 3\}$ receive the singleton and let j receive the empty bundle. Envy-freeness for agent 1 with respect to i requires $1 - p_1 \leq 0 - p_i$, and hence

$$p_1 \geq p_i + 1.$$

Envy-freeness for agent i with respect to j requires $1 - p_i \leq -p_j$, and hence

$$p_i \geq p_j + 1.$$

Combining the two inequalities gives $p_1 \geq p_j + 2$, which is impossible for $p \in \{0, 1\}^3$. Hence no assignment of the additional chore c , even after an arbitrary permutation of the previously allocated bundles, preserves the unit-subsidy guarantee.

Notice that this is an obstruction to the incremental construction, not to the instance itself. The complete allocation $(\{a\}, \{b\}, \{c\})$ is envy-free without subsidies: agent 1 evaluates every singleton at cost 0, while agents 2 and 3 evaluate every singleton at cost 1.

B Unit subsidies are incompatible with Pareto optimality

Recall that, in cost form, a complete allocation $\mathcal{B}$ *Pareto dominates* a complete allocation $\mathcal{A}$ if $c_i(\mathcal{B}_i) \leq c_i(\mathcal{A}_i)$ for every $i \in N$, with strict inequality for at least one, and that $\mathcal{A}$ is *Pareto optimal* if no complete allocation dominates it. The quantifier ranges over complete allocations: for chores, withholding an item weakly benefits every agent, so allowing incomplete allocations would leave no allocation Pareto optimal.

Example 40 (Unit subsidies and Pareto optimality are incompatible). Consider two agents $N = \{1, 2\}$ and three chores $M = \{a, b, c\}$, with both agents sharing the cost function

$$c_1(S) = c_2(S) = \min\{|S|, 2\}, \quad \text{that is} \quad \begin{array}{c|cccc} |S| & 0 & 1 & 2 & 3 \\ \hline c_i(S) & 0 & 1 & 2 & 2 \end{array}$$

Every marginal cost belongs to $\{0, 1\}$, so this is a dichotomous cost function; it is not additive, since $c_i(\{a, b, c\}) = 2 \neq 3$.

Because the two agents share one cost function, envy-freeness of $(\mathcal{A}, p)$ reads $c(A_1) - p_1 = c(A_2) - p_2$, so the subsidy difference is pinned to the cost difference and the minimal subsidy is $p_i^* = \max\{0, c(A_i) - c(A_j)\}$. The eight complete allocations are therefore as follows.

A_1	A_2	costs	p^*	$p^* \leq 1$	PO	dominated by
$\{a, b, c\}$	$\emptyset$	(2, 0)	(2, 0)	no	yes	—
$\{a, b\}$	$\{c\}$	(2, 1)	(1, 0)	yes	no	$(\{a, b, c\}, \emptyset)$
$\{a, c\}$	$\{b\}$	(2, 1)	(1, 0)	yes	no	$(\{a, b, c\}, \emptyset)$
$\{b, c\}$	$\{a\}$	(2, 1)	(1, 0)	yes	no	$(\{a, b, c\}, \emptyset)$
$\{a\}$	$\{b, c\}$	(1, 2)	(0, 1)	yes	no	$(\emptyset, \{a, b, c\})$
$\{b\}$	$\{a, c\}$	(1, 2)	(0, 1)	yes	no	$(\emptyset, \{a, b, c\})$
$\{c\}$	$\{a, b\}$	(1, 2)	(0, 1)	yes	no	$(\emptyset, \{a, b, c\})$
$\emptyset$	$\{a, b, c\}$	(0, 2)	(0, 2)	no	yes	—

No allocation is both Pareto optimal and envy-free with a subsidy of at most one per agent.

The two columns fail for the same reason. The cost function saturates at $|S| = 2$, so the third chore is free to an agent who already holds two: concentrating all three chores on one agent costs 2 in total, whereas every split costs 3, which is why exactly the two concentrated allocations are Pareto optimal and every split is dominated by one of them. But that same saturation makes the loaded agent's cost exceed the empty agent's by 2, so each concentrated allocation needs a subsidy of 2.

Consequently the unit-subsidy guarantee of Theorem 22 cannot be strengthened to deliver a Pareto optimal allocation, and enforcing it costs efficiency: on this instance the cheapest admissible allocation has total cost 3 against 2 for the cheapest Pareto optimal one.

C The One-Good Extension when the Bundles Contain Chores

This appendix restates the one-good extension argument of Barman et al. [BKNS22] in a self-contained way and checks that each step uses only the two facts named in the proof of Lemma 25: the valuations are integer valued, and the good being inserted has marginal value in $\{0, 1\}$ for every agent on every bundle. In particular, no step uses the sign of the items already allocated.

Proof of Lemma 25 For an allocation $Y = (Y_1, \dots, Y_n)$, let G_Y denote its weighted envy graph, with edge weights

$$w_Y(i, j) := v_i(Y_j) - v_i(Y_i).$$

We use two standard characterizations of envy-freenability [HS19, BKNS22] (Theorems 9 and 10). First, Y is envy-freeable if and only if it maximizes

$$\sum_{i \in N} v_i(Y_{\sigma(i)})$$

over all permutations σ of its bundles. Equivalently, G_Y contains no positive-weight directed cycle. Second, if Y is envy-freeable, then its pointwise-minimal supporting subsidy for agent i is the maximum weight of a directed path starting at i in G_Y .

We may assume without loss of generality that p is the pointwise-minimal subsidy vector supporting A . Indeed, starting from the subsidy vector in the statement, replace it by the pointwise-minimal one. This can only decrease the subsidies componentwise. Since the valuations are integer-valued, all edge weights in G_A , and therefore all pointwise-minimal subsidies, are integers. Consequently, the resulting vector still belongs to $\{0, 1\}^n$.

We first prove two properties of adding g that will be used throughout the proof.

Claim 1. Let Y be an envy-freeable allocation of M' , and suppose that for some agent x ,

$$v_x(Y_x \cup \{g\}) - v_x(Y_x) = 1.$$

Then

$$Z = (Y_1, \dots, Y_x \cup \{g\}, \dots, Y_n)$$

is envy-freeable.

Proof of Claim 1. Write

$$\text{SW}(Y) := \sum_{i \in N} v_i(Y_i).$$

Since the marginal value of g for x on Y_x is one,

$$\text{SW}(Z) = \text{SW}(Y) + 1.$$

Consider any permutation σ of the bundles of Z . Let a be the agent who receives the bundle containing g under this permutation. Removing g from that bundle leaves a permutation of the bundles of Y . Since Y is envy-freeable, the welfare of this reassignment is at most $\text{SW}(Y)$. Moreover,

$$v_a(S \cup \{g\}) - v_a(S) \leq 1$$

for every $S \subseteq M'$. Hence

$$\text{SW}(Z_\sigma) \leq \text{SW}(Y) + 1 = \text{SW}(Z).$$

Thus Z maximizes welfare over all permutations of its bundles and is therefore envy-freeable. $\square$

Claim 2. Let Y be an allocation and let

$$Z = (Y_1, \dots, Y_x \cup \{g\}, \dots, Y_n).$$

Then

$$w_Z(i, j) = w_Y(i, j) \quad \text{for all } i, j \neq x,$$

while

$$w_Z(x, j) \leq w_Y(x, j) \quad \text{and} \quad w_Z(i, x) \leq w_Y(i, x) + 1.$$

Consequently, for every simple directed path P ,

$$w_Z(P) \leq w_Y(P) + 1. \tag{11}$$

Proof of Claim 2. Only the bundle of x changes. Hence, edges not incident to x are unchanged. For every $j \neq x$,

$$\begin{aligned} w_Z(x, j) &= v_x(Y_j) - v_x(Y_x \cup \{g\}) \\ &= w_Y(x, j) - (v_x(Y_x \cup \{g\}) - v_x(Y_x)) \leq w_Y(x, j), \end{aligned}$$

because the marginal value of g is nonnegative. Similarly, for $i \neq x$,

$$\begin{aligned} w_Z(i, x) &= v_i(Y_x \cup \{g\}) - v_i(Y_i) \\ &= w_Y(i, x) + (v_i(Y_x \cup \{g\}) - v_i(Y_x)) \leq w_Y(i, x) + 1. \end{aligned}$$

A simple path enters x at most once. Its weight can therefore increase by at most one, proving (11). $\square$

We now follow the extendable/non-extendable distinction of Barman et al. [BKNS22]. For a subsidy vector q , recall that

$$M(q) := \{i \in N : q_i \geq q_j \text{ for every } j \in N\}.$$

Call (A, p) *extendable with g* if there exists a permutation σ such that, writing

$$B_i := A_{\sigma(i)} \quad \text{and} \quad q_i := p_{\sigma(i)},$$

the pair (B, q) is envy-free and there is some $\kappa \in M(q)$ satisfying

$$v_\kappa(B_\kappa \cup \{g\}) - v_\kappa(B_\kappa) = 1. \tag{12}$$

We will also use the following elementary permutation property. If (A, p) is envy-free and $B = A_\sigma$ is envy-freeable, then (B, p_σ) is envy-free. To see this, the envy-freeness of (A, p) gives

$$v_i(A_i) + p_i \geq v_i(A_{\sigma(i)}) + p_{\sigma(i)} \quad \text{for every } i.$$

Summing these inequalities and using the fact that both A and B are envy-freeable shows that equality must hold for every i . The envy-freeness inequalities for (B, p_σ) then follow directly. See Lemma 3 in Barman et al. [BKNS22] for a detailed argument.

Case 1: (A, p) is extendable with g .

Let σ and κ satisfy the above conditions, and define

$$C = (B_1, \dots, B_\kappa \cup \{g\}, \dots, B_n).$$

By Claim 1, C is envy-freeable.

It remains to show that C can be supported by subsidies in $\{0, 1\}^n$. Since q_κ is maximal and $q \in \{0, 1\}^n$, there are two cases.

If $q_\kappa = 0$, then $q_i = 0$ for all $i \in N$. Thus, every directed path in G_B has weight at most zero. By Claim 2, every directed path in G_C has weight at most one. Hence, the pointwise-minimal supporting subsidies for C are at most one. They are nonnegative integers, so they belong to $\{0, 1\}^n$.

Suppose next that $q_\kappa = 1$. Define

$$\widehat{q}_\kappa := 0, \quad \widehat{q}_i := q_i \quad (i \neq \kappa).$$

We claim that $(C, \widehat{q})$ is envy-free. For $i, j \neq \kappa$, nothing changes from (B, q) . For agent κ , using (12),

$$v_\kappa(C_\kappa) + \widehat{q}_\kappa = v_\kappa(B_\kappa) + 1 = v_\kappa(B_\kappa) + q_\kappa \geq v_\kappa(B_j) + q_j.$$

Finally, for every $i \neq \kappa$,

$$\begin{aligned} v_i(C_i) + \widehat{q}_i &= v_i(B_i) + q_i \\ &\geq v_i(B_\kappa) + q_\kappa \\ &= v_i(B_\kappa) + 1 \\ &\geq v_i(B_\kappa \cup \{g\}) = v_i(C_\kappa) + \widehat{q}_\kappa, \end{aligned}$$

where the last inequality follows because the marginal value of g is at most one. Thus $\widehat{q} \in \{0, 1\}^n$ supports C .

Hence, the desired extension exists in the extendable case.

Case 2: (A, p) is not extendable with g .

We use the `FINDSINK` procedure of Barman et al. [BKNS22], and verify that its proof remains valid in the present setting.

For $s \in N$, write

$$X^s := (A_1, \dots, A_s \cup \{g\}, \dots, A_n).$$

Start with an arbitrary $s_0 \in M(p)$. Whenever X^{s_τ} is envy-freeable, let ϕ^τ denote its pointwise-minimal supporting subsidy vector. If every component of ϕ^τ is strictly smaller than 2, terminate. Otherwise, choose an agent $s_{\tau+1}$ with

$$\phi_{s_{\tau+1}}^\tau \geq 2$$

and continue with $X^{s_{\tau+1}}$.

We first establish the following:

$$s_\tau \in M(p) \quad \text{and} \quad X^{s_\tau} \text{ is envy-freeable} \tag{13}$$

for every agent considered by the procedure.

For $\tau = 0$, the first part follows from the choice of s_0 . Suppose, towards a contradiction, that X^{s_0} is not envy-freeable. By the welfare characterization, there exists a permutation of its bundles whose welfare is strictly larger. Since all valuations are integer-valued, the improvement is at least one.

Let B be the corresponding permutation of the bundles of A , obtained by deleting g from the permuted bundle containing it, and let k be the agent who receives the bundle A_{s_0} in B . Since the marginal value of g is nonnegative,

$$\text{SW}(X^{s_0}) \geq \text{SW}(A).$$

Therefore,

$$\text{SW}(B) + (v_k(B_k \cup \{g\}) - v_k(B_k)) \geq \text{SW}(A) + 1. \tag{14}$$

On the other hand, A is envy-freeable, and hence

$$\text{SW}(B) \leq \text{SW}(A).$$

The marginal term in (14) is at most one. Consequently, both inequalities must be tight:

$$\text{SW}(B) = \text{SW}(A)$$

and

$$v_k(B_k \cup \{g\}) - v_k(B_k) = 1.$$

Thus B is envy-freeable. By the permutation property above, if q is obtained by permuting p in the same way, then (B, q) is envy-free. Moreover, $B_k = A_{s_0}$ and $s_0 \in M(p)$, so $k \in M(q)$. Hence (A, p) is extendable with g , a contradiction. Therefore X^{s_0} is envy-freeable.

Now suppose (13) holds for s_τ and the procedure selects $s_{\tau+1}$ because $\phi_{s_{\tau+1}}^\tau \geq 2$. We claim that $s_{\tau+1} \in M(p)$. If not, then the maximal subsidy in p is one and $p_{s_{\tau+1}} = 0$. Since p is pointwise-minimal, every path starting from $s_{\tau+1}$ in G_A has weight at most zero. By Claim 2, every path starting from $s_{\tau+1}$ in $G_{X^{s_\tau}}$ has weight at most one. Hence $\phi_{s_{\tau+1}}^\tau \leq 1$, a contradiction. Thus $s_{\tau+1} \in M(p)$. The same argument used for s_0 then shows that $X^{s_{\tau+1}}$ is envy-freeable. This proves (13).

Note also that $s_{\tau+1} \neq s_\tau$. Indeed, a simple path starting at s_τ never enters s_τ , so by Claim 2 its weight in $G_{X^{s_\tau}}$ is at most its weight in G_A , which is at most $p_{s_\tau} \leq 1$. Hence $\phi_{s_\tau}^\tau \leq 1$, and s_τ is never selected immediately after itself.

It remains to show that the procedure terminates. Suppose otherwise that some agent is selected twice. Consider two consecutive occurrences of a repeated agent and relabel the corresponding segment as

$$s_0, s_1, \dots, s_T, \quad s_T = s_0,$$

with $s_0, \dots, s_{T-1}$ distinct. By the preceding paragraph, $T \geq 2$.

For every $\tau = 1, \dots, T$, agent s_τ is selected because its pointwise-minimal subsidy for $X^{s_{\tau-1}}$ is at least two. Hence, there exists a simple directed path Q_τ starting at s_τ in $G_{X^{s_{\tau-1}}}$ such that

$$w_{X^{s_{\tau-1}}}(Q_\tau) \geq 2. \quad (15)$$

The path Q_τ must contain $s_{\tau-1}$. Otherwise, none of its edge weights differs from the corresponding edge weights in G_A , whereas every path starting at s_τ in G_A has weight at most 1, contradicting (15).

Let P_τ be the initial segment of Q_τ from s_τ to $s_{\tau-1}$. By Claim 2,

$$w_A(Q_\tau) \geq w_{X^{s_{\tau-1}}}(Q_\tau) - 1 \geq 1. \quad (16)$$

The part of Q_τ following P_τ is a path starting at $s_{\tau-1}$ in G_A , and therefore has weight at most $p_{s_{\tau-1}} \leq 1$. It follows from (16) that

$$w_A(P_\tau) \geq 0. \quad (17)$$

Now concatenate the paths $P_T, P_{T-1}, \dots, P_1$. Since $s_T = s_0$, this concatenation starts and ends at s_0 . If some vertex is repeated in the concatenation, split the resulting closed sequence of edges at repeated vertices. In this way, it can be decomposed into directed cycles of G_A . By (17), the total weight of the concatenated sequence is nonnegative. On the other hand, since A is envy-freeable, G_A contains no positive-weight directed cycle. Therefore, every cycle arising in the above decomposition has weight at most zero, while their total weight is nonnegative. It follows that the total weight is zero, and consequently, every cycle in the decomposition has weight zero.

We now derive a contradiction to non-extendability. Since s_1 was selected after considering X^{s_0} , there is a path Q_1 satisfying (15). Its weight in G_A is at most $p_{s_1} \leq 1$, while Claim 2 says that its weight can increase by at most one after g is added to A_{s_0} . Hence

$$w_{X^{s_0}}(Q_1) = 2, \quad w_A(Q_1) = 1,$$

and the weight of Q_1 increases by exactly one. Only an edge entering s_0 can increase when g is added to A_{s_0} . Edges leaving s_0 can only decrease. Since $s_1 \neq s_0$ and Q_1 starts at s_1 and contains s_0 , the agent s_0 has a predecessor k on Q_1 , and

$$v_k(A_{s_0} \cup \{g\}) - v_k(A_{s_0}) = 1. \quad (18)$$

Since the edge (k, s_0) belongs to P_1 , it belongs to one of the zero-weight directed cycles obtained above. Let C be such a cycle. Cyclically permuting the bundles of A along C preserves social welfare, because the change in welfare is exactly $w_A(C) = 0$. Let σ denote the resulting permutation and let $B = A_\sigma$. Hence

$$\text{SW}(B) = \text{SW}(A),$$

so B is envy-freeable. By the permutation property, (B, p_σ) is envy-free.

Furthermore, the cycle contains (k, s_0) , so agent k receives bundle A_{s_0} in B . Thus (18) gives

$$v_k(B_k \cup \{g\}) - v_k(B_k) = 1.$$

Since $s_0 \in M(p)$, agent k belongs to $M(p_\sigma)$. Consequently, σ and k witness that (A, p) is extendable with g , contradicting the assumption of Case 2.

Therefore, no agent is selected twice, and the procedure terminates after at most n selections. At termination, the current allocation X^s is envy-freeable by (13), and every component of its pointwise-minimal subsidy vector is strictly smaller than 2. These subsidies are nonnegative integers, and hence they belong to $\{0, 1\}^n$. This proves the existence of the desired allocation A' and subsidy vector p' in the non-extendable case.

Finally, all steps are polynomial-time. Pointwise-minimal subsidies can be computed from maximum-weight paths in an envy-freeable allocation. Extendability can be tested by the `EXTEND` subroutine of Barman et al. [BKNS22]: for each candidate recipient and most-subsidized bundle, it solves a maximum-weight bipartite matching problem. Its correctness uses only the welfare characterization of envy-freeability and the values of the existing bundles, and is therefore unchanged here. In the non-extendable case, the procedure above considers at most n agents. Thus, the entire one-good extension can be carried out in polynomial time using value-oracle access. $\square$